

Exhaust Gas Back Pressure Report

Report Generated: 2026-06-08 15:49:43

Exhaust Gas Sources

Source	Mass Flow (kg/s)	Temp (°C)	Max BP (Pa)	Roughness
Main Engine	80.651	240	3000	Steel Welded
Auxiliary Engine 2	7.395	365	3000	Steel Welded
Auxiliary Engine 3	7.395	365	3000	Steel Welded

System Summary (Combined)

Total Mass Flow	95.441 kg/s
Avg. Temperature	259.4 °C
System Max BP Limit	3000 Pa
Design Roughness	Steel Welded

Overall Assessment

Total Back Pressure	2063.15 Pa
Total Back Pressure	210.382 mmWC
Compliance Status	PASSED
Safety Margin	31.23%

Pipeline Elements Breakdown

#	Element	Velocity (m/s)	ξ	ΔP (Pa)
1	Pipe	71.60	0.007	11.9
2	Wye Branch	71.60	0.200	339.9
3	Pipe	71.60	0.007	11.9
4	Custom	71.60	1.000	1699.4